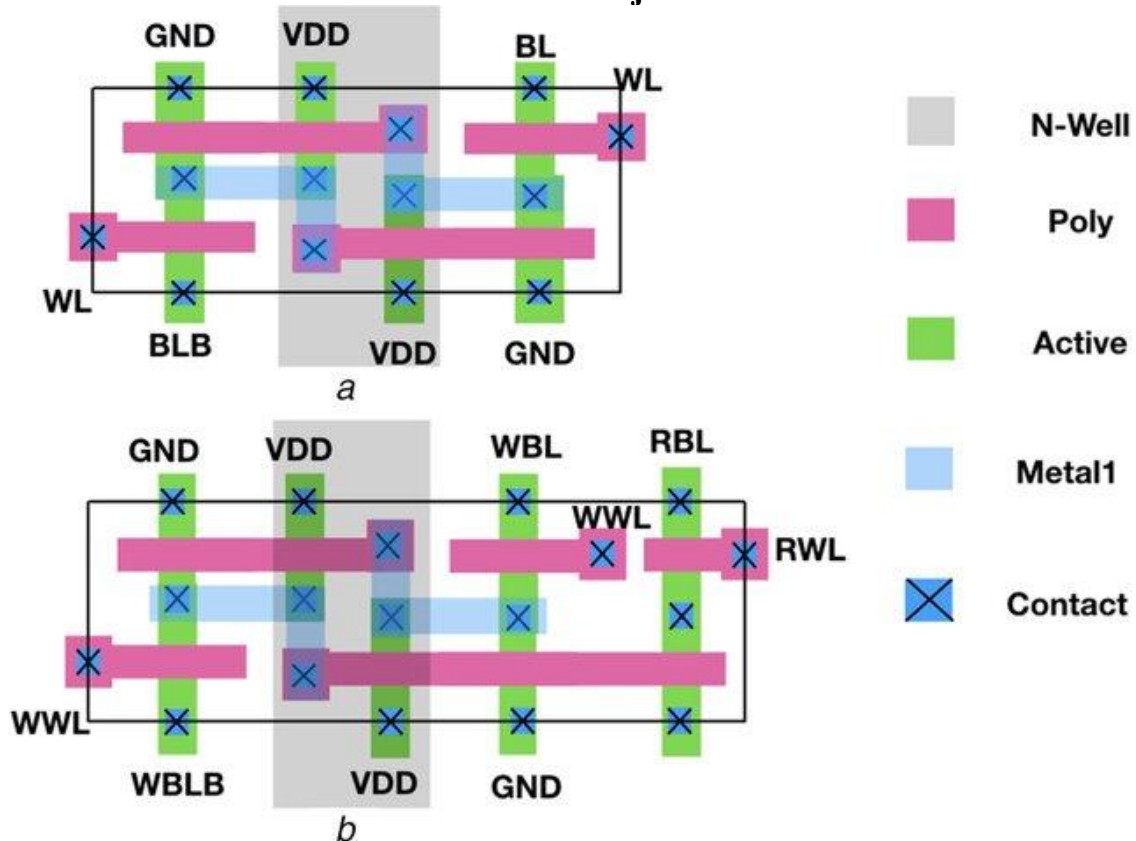




Course Project



You are given the symbolic layout shown in the figure. It is required to draw the extracted schematic and then redraw its layout using any suitable software. The layout should pass the LVS and DRC tests.

You can use any suitable software for schematic and layout:

- 1- Cadence
- 2- Microwind <https://www.microwind.net/>
- 3- Electric <http://www.staticfreesoft.com>

(It is recommended to begin using Electric if you have no background on any simulation tools)

Deliverables:

- A documentation for the project reporting the schematic, layout drawn, DRC results, LVS results etc.
- A .zip file containing all your project files (schematic and layout).

The report should be uploaded to the LMS before the **8th of May at 11:59 PM**.

General Instructions:

- Use any available process design kit (PDK)
- Use any sizing you see appropriate for the circuits.

Evaluation Criteria

1	Layout	30%
2	Schematics	30%
3	DRC	20%
4	LVS	20%

Bonus (5%)

Extract the parasitics of the layout and do post-layout simulations showing the difference between figures a and b.

Academic Misconduct

The University defines Academic Misconduct as ‘any case of deliberate, premeditated cheating, collusion, plagiarism or falsification of information, in an attempt to deceive and gain an unfair advantage in assessment’. This includes attempting to gain marks as part of a team without making a contribution. The department takes Academic Misconduct very seriously and any suspected cases will be investigated through the University’s standard policy. If you are found guilty, you may be expelled from the University with no award.

It is your responsibility to ensure that you understand what constitutes Academic Misconduct and to ensure that you do not break the rules. If you are unclear about what is required, please ask.